

AP Statistics FRQ Practice Exam 09

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AP Statistics FRQ Practice Exam 9

APSTAT-FRQ9-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: Unit 3, Inference for Categorical Data: Proportions

FRQ Type: Formulate Questions and Collect Data

Prompt:

A county with 48,000 registered voters is considering whether to lease public land for a solar farm. The county places an open poll on its website and advertises the poll through its social-media accounts. Any registered voter may choose to log in and respond once. The results are shown.

Response	Number
Support	2,304
Oppose	896
Total	3,200

- (a) Identify the population, sample, and population parameter of interest.
- (b) Identify the sampling method. Explain two distinct ways the method could produce a biased estimate of countywide support.
- (c) Calculate the respondent sample proportion who support the proposal and interpret it. Explain why the large number of responses does not by itself eliminate the bias.
- (d) State a statistical question the county wants to answer. Explain why a standard margin of error based on random sampling would not be justified for these poll results.
- (e) Describe a probability-sampling redesign using the voter registry. Include one step for reducing or assessing nonresponse.

Scoring Guide:

Total Points: 10

(a) 2 points

· Point 1: Identifies the population as all 48,000 registered voters and the sample as the 3,200 voters who chose to respond to the online poll.

- Point 2: Identifies p as the true proportion of all registered voters in the county who support leasing the public land for the solar farm.

(b) 2 points

- Point 1: Identifies the method as a voluntary-response sample rather than a random sample.
- Point 2: Explains two distinct bias mechanisms, such as people with strong opinions being more likely to respond and voters who follow county social media or regularly use the website being overrepresented.

(c) 2 points

- Point 1: Calculates $\hat{p} = 2,304/3,200 = 0.720$ and interprets it as 72.0% support among the poll respondents.
- Point 2: Explains that increasing the number of self-selected responses can reduce random variability around the respondent-group value but does not correct systematic self-selection or coverage bias.

(d) 2 points

- Point 1: States an appropriate statistical question, such as, "What proportion of all registered county voters support leasing the land for the solar farm?"
- Point 2: Explains that standard margin-of-error formulas require a probability sample or comparable random mechanism; the voluntary sample provides no known selection probabilities and may not represent the population.

(e) 2 points

- Point 1: Proposes selecting an SRS of voter records, or another clearly described probability sample, and contacting only the selected voters with a neutral question.
- Point 2: Includes a relevant nonresponse step, such as repeated contacts, multiple response modes, or intensive follow-up of a random subsample of nonrespondents to compare their views with respondents' views.

Model Guidance:

The 3,200 respondents are numerous but self-selected. The resulting 72% is a description of that respondent group, not a defensible estimate with a random-sampling margin of error for all 48,000 registered voters. Strongly interested voters and voters connected to the county's online channels can have much higher response probabilities than others.

A better design begins with the voter registry and selects records by chance. The county should use neutral wording, contact the selected voters directly, and follow up with initial nonrespondents through more than one mode when possible. Random selection addresses representation; follow-up addresses the possibility that selected nonrespondents differ from respondents.

Expected Points:

10

APSTAT-FRQ9-002

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: None

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

Twenty-four volunteers complete the same visual-search task using one of two screen interfaces. The volunteers are randomly assigned, 12 to each interface. Completion times, in seconds, are shown in increasing order.

Interface A: 12, 14, 15, 15, 16, 17, 18, 19, 20, 22, 24, 31

Interface B: 13, 13, 14, 16, 16, 16, 17, 17, 18, 18, 19, 20

(a) On paper, construct a back-to-back stemplot for the two distributions using tens digits as stems and ones digits as leaves. Include a key and label both sides.

(b) Compare the distributions with respect to shape, center, spread, and unusual features.

(c) Calculate the mean and median for each interface. Use the results to compare typical completion time.

(d) Calculate Q_1 , Q_3 , and the IQR for each interface. Apply the $1.5 \times \text{IQR}$ rule to determine whether either distribution has a potential outlier.

(e) A designer claims that Interface B causes faster and more consistent completion for volunteers like those studied. Evaluate the claim using the numerical evidence and the study design.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Constructs a correct back-to-back stemplot with stems 1, 2, and 3 and places every time on the correct side in numerical order.
- Point 2: Labels Interface A and Interface B and includes a key such as “1 | 7 represents 17 seconds.”

(b) 2 points

- Point 1: Describes Interface A as having a longer upper tail with the isolated high value 31, while Interface B is more compact and approximately symmetric or only mildly skewed.
- Point 2: States that Interface B has a slightly lower center and substantially less spread than Interface A, using the display in context.

(c) 2 points

- Point 1: Calculates $\bar{x}_A = 223/12 \approx 18.58$ seconds and $\text{median}_A = 17.5$ seconds.
- Point 2: Calculates $\bar{x}_B = 197/12 \approx 16.42$ seconds and $\text{median}_B = 16.5$ seconds, supporting a lower typical time for Interface B.

(d) 2 points

- Point 1: For Interface A, calculates $Q_1 = 15$, $Q_3 = 21$, $\text{IQR} = 6$, fences 6 and 30, and identifies 31 as a potential high outlier.
- Point 2: For Interface B, calculates $Q_1 = 15$, $Q_3 = 18$, $\text{IQR} = 3$, fences 10.5 and 22.5, and finds no potential outlier.

(e) 2 points

- Point 1: Supports faster and more consistent performance for the studied volunteers using B's lower mean and median and its smaller IQR, 3 seconds versus 6 seconds.
- Point 2: Explains that random assignment supports a causal comparison for the volunteers, but volunteer recruitment does not justify generalization to all potential users without a representative sample.

Model Guidance:

The back-to-back stemplot should show that most times overlap in the teens, but Interface A extends through the twenties to 31 seconds. Its mean exceeds its median because of the longer upper side. Interface B has a mean and median both near 16.5 seconds and a much smaller IQR.

For A, the upper fence is 30 seconds, so 31 is flagged. B has no flagged value. Both measures of center favor B, while the IQR comparison supports greater consistency. Because the volunteers were randomly assigned to interfaces, the observed treatment difference can be interpreted causally for similar experimental participants. Their volunteer status limits population generalization.

Expected Points:

10

APSTAT-FRQ9-003

Section: FRQ

Type: frq

Primary Unit: Unit 4, Inference for Quantitative Data: Means

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

A facilities department randomly selects 16 workshops from the 240 workshops it manages. Sound level is measured in each selected workshop during a standardized operating period. Acoustic panels are then installed, and sound level is measured again under the same stated operating protocol one month later.

For each workshop, define d = after-installation sound level – before-installation sound level, in decibels. The differences have

$$n = 16$$

$$\bar{x}_d = -3.8 \text{ dB}$$

$$s_d = 4.5 \text{ dB}$$

A dotplot of the 16 differences is roughly symmetric with no apparent outliers. At the $\alpha = 0.01$ significance level, test whether the true mean sound level was lower after installation.

(a) Define the population parameter and state the hypotheses.

(b) Identify the appropriate inference procedure and check its conditions.

(c) Calculate the standard error, test statistic, degrees of freedom, and p-value.

(d) State a conclusion in context using the $\alpha = 0.01$ significance level.

(e) Explain one limitation of attributing the observed before-after change entirely to the acoustic panels.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Defines μ_d as the true mean after-minus-before sound-level difference for all 240 workshops managed by the department under the stated protocol.
- Point 2: States $H_0: \mu_d = 0$ dB and $H_a: \mu_d < 0$ dB.

(b) 3 points

- Point 1: Selects a paired t test, equivalently a one-sample t test on the workshop differences.
- Point 2: Cites the random sample of workshops, the pairing of before and after measurements within workshops, and verifies $16 \leq 0.10(240) = 24$, supporting independence of the sampled workshop differences.
- Point 3: Uses the roughly symmetric, outlier-free distribution of differences to justify a t procedure for the small sample.

(c) 3 points

- Point 1: Calculates $SE = sd/\sqrt{n} = 4.5/\sqrt{16} = 1.125$ dB.
- Point 2: Calculates $t = (-3.8 - 0)/1.125 \approx -3.38$ with $df = 16 - 1 = 15$.
- Point 3: Calculates the left-tail p-value $P(T_{15} \leq -3.38) \approx 0.0021$.

(d) 1 point

- Point 1: Because $0.0021 < 0.01$, rejects H_0 and concludes that the data provide convincing evidence that the true mean sound level was lower after installation for the department's workshops.

(e) 1 point

- Point 1: Explains that the before-after design has no concurrent control group, so changes in equipment, workload, staffing, or other time-related factors could contribute to the decrease; pairing alone does not isolate the panels' causal effect.

Model Guidance:

The same workshop is measured twice, so analyze the 16 after-minus-before differences. Their mean is negative, matching the directional alternative. The workshops were randomly sampled, 16 is no more than 10% of 240, and the differences are roughly symmetric without outliers, supporting a paired t procedure.

The standard error is 1.125 dB and $t \approx -3.38$ with 15 degrees of freedom, giving $p \approx 0.0021$. This is below 0.01, so there is evidence of a negative population mean difference. The result documents before-after change, but without an untreated comparison group it does not prove that the panels alone caused the change.

Expected Points:

10

APSTAT-FRQ9-004

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

A laboratory has four coded sample vials and four numbered refrigerated slots. Each vial belongs in exactly one matching slot. During a barcode-system outage, a technician randomly places one vial in each slot, with all $4! = 24$ assignments equally likely. Let X be the number of vials placed in their correct slots.

x , correct placements	0	1	2	3	4
Number of assignments	9	8	6	0	1

- Construct the probability distribution of X and verify that it is valid. Explain why $X = 3$ is impossible.
- Calculate and interpret $E(X)$. Determine the median and mode of X , and explain what each measure says about the distribution.
- Calculate $P(X \geq 1)$ and $P(X \geq 2 \mid X \geq 1)$. Interpret the conditional probability.
- Describe a simulation using four cards or random digits that would reproduce one random assignment per trial and estimate $P(X \geq 2 \mid X \geq 1)$.
- A supervisor says, “Exactly one vial is the most likely outcome because the expected number correct is one.” Evaluate both parts of the reasoning using the distribution.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Gives probabilities $9/24 = 0.375$, $8/24 \approx 0.333$, $6/24 = 0.250$, 0, and $1/24 \approx 0.0417$ for $X = 0, 1, 2, 3$, and 4, and verifies that they sum to 1.
- Point 2: Explains that if three vials are in their correct slots, the only remaining vial and slot must also match, so exactly three correct placements cannot occur.

(b) 2 points

- Point 1: Calculates $E(X) = [0(9) + 1(8) + 2(6) + 4(1)]/24 = 1$ and interprets this as a long-run average of one correct placement per random assignment.
- Point 2: Identifies the median as 1 and the mode as 0, explaining that at least half the assignments have $X \leq 1$ and at least half have $X \geq 1$, while $X = 0$ has the largest single probability.

(c) 2 points

- Point 1: Calculates $P(X \geq 1) = (8 + 6 + 1)/24 = 15/24 = 0.625$.

· Point 2: Calculates $P(X \geq 2 | X \geq 1) = (6 + 1)/(8 + 6 + 1) = 7/15 \approx 0.467$ and interprets it as the proportion with at least two correct among assignments having at least one correct.

(d) 2 points

· Point 1: Describes randomly shuffling four distinctly labeled vial cards into four fixed slot positions, counting position-label matches, and recording whether $X \geq 1$ and $X \geq 2$ for each trial.

· Point 2: Repeats many trials and estimates the conditional probability as the number of trials with $X \geq 2$ divided by the number of trials with $X \geq 1$, excluding trials with $X = 0$ from the denominator.

(e) 2 points

· Point 1: Rejects the claim that exactly one is most likely because $P(X = 0) = 0.375$ exceeds $P(X = 1) = 0.333$.

· Point 2: Explains that an expected value is a long-run balance point, not necessarily the mode or the outcome most likely in one assignment; therefore, $E(X) = 1$ does not justify the supervisor's conclusion.

Model Guidance:

Divide each assignment count by 24 to form the distribution. Exactly three matches are impossible because the fourth placement would then be forced to match. The expected value is 1, the median is 1, and the mode is 0; these measures answer different questions and need not coincide.

At least one vial is correct in 15 of the 24 assignments. Conditional on that event, 7 of those 15 assignments have at least two correct, giving 7/15. A simulation should create a uniformly random permutation on every trial and use only trials with at least one match in the conditional denominator. The distribution directly refutes the supervisor's attempt to equate expected value with the most probable single outcome.

Expected Points:

10

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ9

· APSTAT-FRQ9-001: Unit 1 primary; Unit 3 secondary; voluntary-response online poll advertised through county web/social channels to 48,000 registered voters, with 2,304 Support and 896 Oppose among 3,200 self-selected respondents; identify population/sample/p, compute respondent $\hat{p} = 0.720$, formulate the county-support question, diagnose strong-opinion and online-coverage bias, explain why large n does not cure bias or justify a random-sampling margin of error, and redesign with a registry SRS plus nonresponse follow-up; targets equating sample size with representativeness.

· APSTAT-FRQ9-002: Unit 1 primary; no secondary unit; randomly assigned screen-interface completion-time datasets for 12 volunteers per interface, analyzed with a labeled back-to-back stemplot; Interface A mean/median 18.58/17.5, Q1/Q3 15/21, IQR 6, high outlier 31, versus Interface B mean/median 16.42/16.5, Q1/Q3 15/18, IQR 3 and no outlier; compare shape, center, spread, support faster/more consistent B for volunteers, and limit generalization from volunteers; targets ignoring spread and outliers when comparing only means.

· APSTAT-FRQ9-003: Unit 4 primary; Unit 1 secondary; paired left-tailed t test for after-minus-before sound level in an SRS of 16 from 240 workshops after acoustic-panel installation, with $\bar{x}_d = -3.8$ dB, $s_d = 4.5$ dB, symmetric outlier-free differences, $t \approx -3.38$, $df = 15$ and $p \approx 0.0021$ at $\alpha = 0.01$; verify the paired/SRS/10%/shape conditions and note absence of a concurrent control group for causation; targets treating paired measurements as independent samples and confusing before-after evidence with isolated treatment causation.

· APSTAT-FRQ9-004: Unit 2 primary; Unit 1 secondary; random permutation of four coded vials into four matching slots with X correct and assignment counts 9, 8, 6, 0, 1 for $X = 0, 1, 2, 3, 4$; construct the exact distribution, explain why $X = 3$ is impossible, compare $E(X) = 1$, median 1, and mode 0, calculate $P(X \geq 1) = 0.625$ and $P(X \geq 2 | X \geq 1) = 7/15$, design a permutation simulation, and reject equating expected value with the most likely outcome.